Can GenAI make the world Disability Inclusive?

Dennie Declercq // 2025 // Agile Testing Days Prompt Forward Hands-on

agiletestingdays.com

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Content

- Introduction: Why Do I Care?
- Experiences from ‘Schol’!
- How to use GenAI for Disability Inclusive Testing
- Thanks // Q&A

Introduction: Why Do I Care?

Disability Spectrum

Difficulty	Permanent	Temporary	Situational
Reading	Blindness	Eye Infection	Aging
Writing	Dyslexia	Stressed	Distracted in Public Transport
Understanding	Down Syndrome	Dementia	In a new country
Daily Life	Autism/ Down Syndrome	A Concussion	A new parent
Professional & Societal Life	Autism, ADHD	Recovering from health problem/ surgery	In a new job

Quality of Life (QoL)

- How much Life Quality does someone have? (appr. 6 models)
- Schalock & Verdugo Model:
  - Most used in the social industry for People with Disabilities
  - 8 Dimensions
- Can be enriched via GenAI (at least):
  - Self-Determination
  - Social Inclusion
  - Material Well-Being

8 Dimensions of QoL
Personal Development
Self-Determination
Interpersonal Relations
Social Inclusion
Rights
Emotional Well-Being
Physical Well-Being
Material Well-Being

Citizen Tech

- Technology that **NEEDS** to be used in private life by Citizens
- Citizens that can't use the applications are excluded
- Examples:
  - Banking & Financial apps;
  - Festival & Concert ticketing;
  - Medical care and insurance communication;
  - Entrance, crowd control, shopping, etc.
- If they are not accessible & inclusive ➔
  - They create a **BARRIER** for citizens;
  - **SEPARATE** and **EXCLUDE** people.

Dimensions of QoL
Self-Determination
Social Inclusion
Material Well-Being

Experiences from ‘Schol’!

Inclusive workshops in an Inclusive shop

- Attendees with:
  - Autism
  - Down Syndrome
  - Non-tech background seniors
  - Parents
- Topics:
  - Social Media
  - Digital Etiquette
  - GenAI to assist yourself

GenAI topics

Topic	Examples
Daily Living	Shopping list // Easy Recipes
Knowledge / Education	What is // Why is // How does it work // Who is
Assistance	Structure // Planning
Helping	Reading text aloud // Translate information

Quality of Life improvements are real

- GenAI for daily life/ assistance (ex. Planning) ➔ increase of Self-Determination
- GenAI for knowledge/ education ➔ increase of Self-Determination, Social Inclusion, Rights
- GenAI for helping ➔ Increase of Self-Determination, Social Inclusion & Material Well-Being

Dimensions of QoL
Self-Determination
Social Inclusion
Material Well-Being

Live demo!

What I Teach @ SCHOL

How to use GenAI for Disability Inclusive Testing

Our story

- “Everyone Wants Christmass”
- An inclusive Christmas stall
- Matching the Dimensions of QoL

Everyone Wants Christmas — Christmas Market Stall

Dimensions of QoL	Features that can increase QoL
Self-Determination	People can choose on their own to visit the stall
Social Inclusion	People know who works at the stall & what to expect
Material Well-Being	People can receive the same information without needing to pay more

Testing Code on Accessibility

- VSCode with GitHub Copilot
  - “Make this page accessible”

A note on factchecking

- Explain me why option Y is more accessible
- Switch to multiple AI models within the same tool
- Use different AI brands and vendors with the same questions

Image description

- Generate ALT-Texts for images
- Works with GPT Vision

Chinese!

- Translate some text to foreign languages

Revert! (To test translation)

- Ask another GenAI tool to translate foreign language to English

Hunting our Competitors + Factchecking

- Bing Copilot in Edge
  - “Is this webpage accessible?”
  - “Test this page on accessibility.”

Live demo!

Everyone Wants Christmas

Warm drinks, sweet cakes, and an inclusive welcome for everyone

Menu About Visit

Welcome to our Christmas market stall

Try our signature hot cocktail — **fuerzangebolwe** — a warm, spiced drink served with love, plus homemade cakes. Our stall is run by people with Down syndrome and we aim to make the experience welcoming and accessible for everyone.

See the menu | Find us

Recap

- **I do care about GenAI for People with disabilities** because:
  - Quality of Life
  - Citizen Tech
- In “Schol” I train people with down syndrome to use GenAI
  - People can do more than they (and you) think
- **You can use GenAI too**:
  - Code testing on Accessibility
  - Alt-Text Generation
  - Translation
  - Do factchecking!!

Thanks! Q&A

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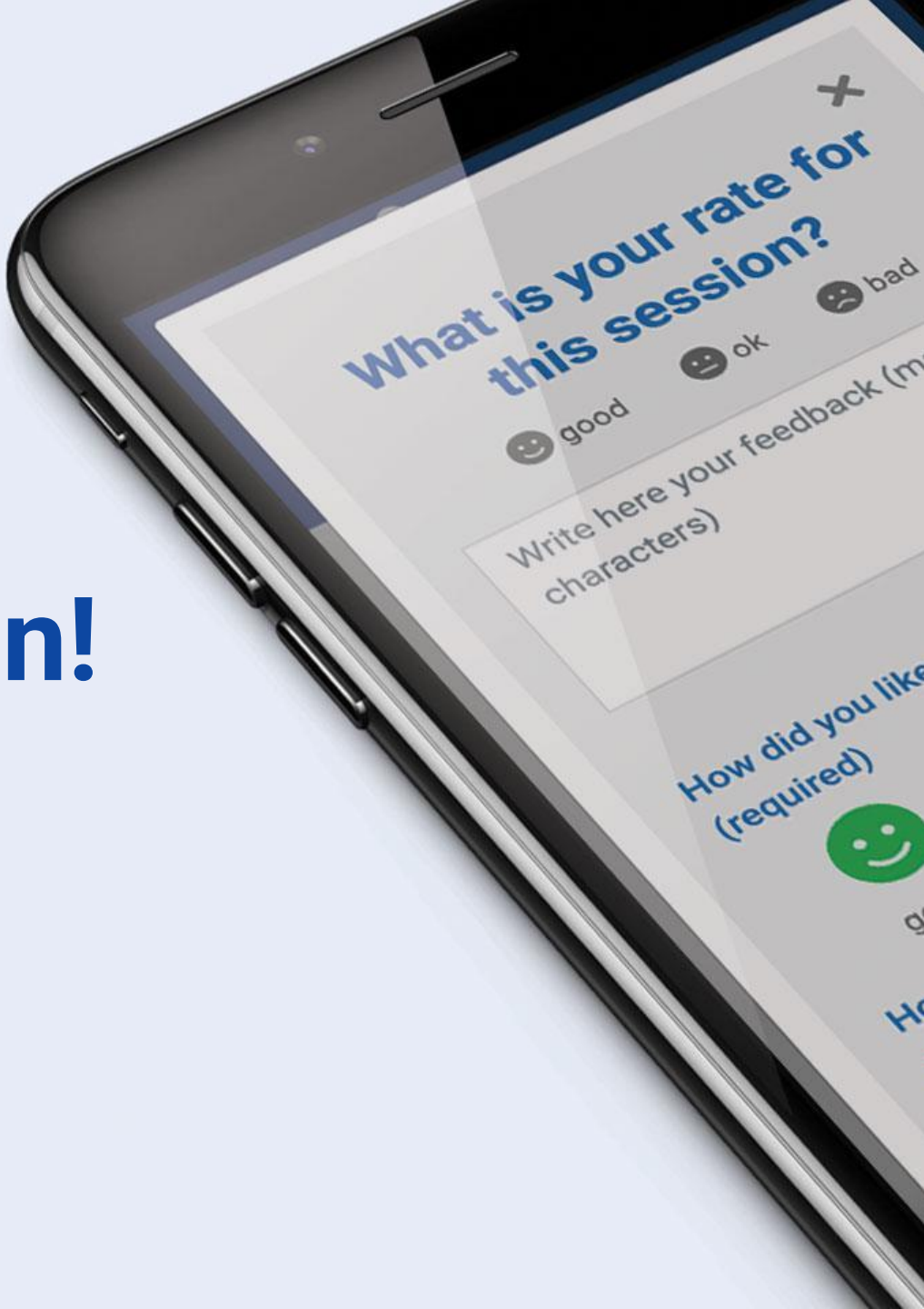
My slides are / will be available for you at:

https://github.com/DDSoftDennie/SlideDecks

Thanks for your attention!

Feedback welcome!

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